1 How to construct the Natural Debt Limit (NDL)

Start:

$$c_t + \frac{b_{t+1}}{R} = y_t + b_t$$

(1) $$b_t = c_t + \frac{b_{t+1}}{R} - y_t$$

(2), (3) $$b_{t+1} = c_{t+1} + \frac{b_{t+2}}{R} - y_{t+1}$$

(4) $$b_t = c_t - y_t + \frac{c_{t+1}}{R} \boxed{+ \frac{b_{t+2}}{R^2}} - \frac{y_{t+1}}{R}$$

(5) Iterate: write down b_{t+2}, substitute ...

(6) Repeat...

(7) Look at term in red:

Assume $\lim_{T \to \infty} R^{-T} b_{t+T} = 0$

This is a Transversality Condition

when $T \to \infty$, you get

$$b_t = \sum_{j=0}^{\infty} R^{-j} \left(c_{t+j} - y_{t+j} \right)$$

Finally, assume $c_t = 0$

$$\Rightarrow \boxed{\hat{b}_t = -\sum_{j=0} R^{-j} y_{t+j}}$$

is the NDL.

Even with very low consumption, $c_t = 0$, the consumer cannot repay more than $\hat{b}_t$.

[The operation we did is called "to solve forward" for b_t]

2. Euler equation

$$L = \sum_{t=0}^{\infty} \left(\beta^t u(c_t) + \lambda_t \left(y_t + b_t - c_t - \frac{b_{t+1}}{R} \right) + \mu_t b_{t+1} \right)$$

$$\frac{\partial L}{\partial c_t} = \beta^t u'(c_t) - \lambda_t = 0$$

$$\frac{\partial L}{\partial b_{t+1}} = -\lambda_t / R + \lambda_{t+1} + \mu_t = 0$$

$$\Rightarrow \frac{\beta^t u'(c_t)}{R} = \beta^{t+1} u'(c_{t+1}) + \mu_t$$

$$u'(c_t) = \beta R \, u'(c_{t+1}) + \frac{\mu_t}{\beta^t} R$$

Assume $\beta R = 1$

$$\Rightarrow u'(c_t) = u'(c_{t+1}) + \frac{\mu_t R}{\beta^t}$$

③ Example 1

$$PV(y) = \sum_{t=0}^{\infty} \beta^t y_t$$

$$= y_h + \beta y_l + \beta^2 y_h + \beta^3 y_l + \beta^4 y_h + \beta^5 y_l + \cdots$$

$$= y_h + \beta y_l + \beta^2 (y_h + \beta y_l) + \beta^4 (y_h + \beta y_l) + \cdots$$

$$= (y_h + \beta y_l)(1 + \beta^2 + \beta^4 + \cdots)$$

$$\text{sum} = 1 + \beta^2 + \beta^4 + \cdots$$

$$\beta^2 \text{ sum} = \beta^2 + \beta^4 + \cdots$$

$$\text{sum} - \beta^2 \text{ sum} = 1$$

$$\text{sum}(1 - \beta^2) = 1$$

$$\text{sum} = \frac{1}{1-\beta^2}$$

$$\Longrightarrow \quad PV(y) = \frac{y_h + \beta y_\ell}{1-\beta^2}$$

4 Constructing sequence of bond holdings

$$c_0 + \frac{b_1}{R} = y_0 + b_0$$

$$\bar{c} + \frac{b_1}{R} = y_h + 0$$

$$b_1 = R(y_h - \bar{c})$$

Substitute $\bar{c}$ . . .

$$c_1 + \frac{b_2}{R} = y_1 + b_1$$

$$\bar{c} + \frac{b_2}{R} = y_\ell + b_1$$

Substitute $\bar{c}$, b_1 . . .